

Elias Bitsch

Mixed Reality for Human-Robot Collaboration | XR Training Systems | User Studies with Industry Partners

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Research Profile

Interactive systems for human-robot collaboration, built end to end and then tested in controlled user studies with industry participants. Mixed reality is the centre of the work, bare-hand authoring and teleoperation across Unity and ROS 2, but the same question is pursued through every channel an operator actually uses: touch interfaces for supervising mobile robots, tangible instruments, and haptic machine controls. Arrives with a working bare-hand teleoperation stack of his own and with study instrumentation that a host lab can point at its own research question rather than rebuild.

Education

UAS Technikum Wien
M.Sc. in Robotics Engineering

Vienna, Austria
Sep 2025 – expected 2027

- Specialisation: Robot Learning
- Master's thesis in preparation on mixed-reality training and skill acquisition in human-robot collaboration

UAS Technikum Wien
B.Sc. in Mechatronics and Robotics

Vienna, Austria
2023 – 2025

- Robotics software in C++, Python, ROS/ROS 2, Gazebo, RViz and OpenCV, deployed in Docker
- Hardware development: PCB design (EasyEDA), CAD and 3D-printed prototypes (SolidWorks, Fusion 360)

HTL Eisenstadt, Higher Technical College
Five-year secondary programme in Mechatronics, GPA 1.86 (1 = best)

Eisenstadt, Austria
2016 – 2021

Publications and Presentations

Bitsch, E., Ovdienko, V., Stix, P., Saliger, A., Angel, N., Orsolits, H. (2026). *MetaMove: Bare-Hand Mixed-Reality Teleoperation of a Collaborative Robot with a Distance-Based Speed-Scaling Safety Layer*. XR-SPro Workshop, IEEE ISMAR 2026 Adjunct Proceedings (ISMAR-Adjunct), Bari, Italy. **Accepted**.

Bitsch, E. (2026). *Human-Robot Interaction for Casting Parts and Defect Annotation*. Live demonstration built for the Applied AI Conference (AAIC), shown in a joint AIT and Fraunhofer session.

Research Experience

AIT Austrian Institute of Technology
Technical Assistant, Center for Technology Experience

Vienna, Austria
Aug 2025 – present

- **Mixed reality (Unity, C#, Meta Quest):** hand-tracked interaction, passthrough, spatial anchors and digital twins, built and deployed as standalone Quest applications
- **Robotics middleware (ROS 2):** MoveIt and MoveIt Servo, Nav2 and behaviour trees, real-time bridges to industrial robot controllers, containerised with Docker
- **Web-based operator interfaces (React, Three.js, Node.js):** 3D supervision dashboards over a Node bridge to ROS 2, usable from a browser without robotics knowledge
- **Embedded feedback hardware (ESP32, IMU, haptics, LEDs):** handheld and wearable instruments from PCB and firmware through to the interaction design

- **Human-robot interaction research:** within-subject designs with controlled failure injection, multimodal feedback across visual, auditory and haptic channels, observational analysis and questionnaires with industry participants
- **Note:** the individual projects are covered by client and institute confidentiality; details on request

UAS Technikum Wien, MetaMove project

Semester project, first author of the resulting publication

Vienna, Austria

2026

- **Bare-hand XR teleoperation:** of an ABB GoFa collaborative robot, Unity to ros-tcp-connector to ROS 2 with MoveIt Servo, an EGM bridge and OmniCore at 250 Hz
- **Safety layer:** distance-based speed scaling that reduces robot velocity as the operator approaches, plus a hardware-free RobotStudio simulation path
- **Outcome:** accepted as a short paper at the XR-SPro workshop, IEEE ISMAR 2026; because the simulation path needs no robot, the stack can be stood up in a new lab before hardware time is available

Professional Experience

Elektrobit Austria GmbH

Software Developer (internship)

Vienna, Austria

Feb – Jun 2025

- **Automotive middleware:** designed and implemented an extension of the SOME/IP protocol that carries and honours quality-of-service parameters in distributed AUTOSAR Classic systems, so that services are discovered and prioritised across ECUs
- **Outcome:** deployed on automotive test hardware and tested successfully

Mars Incorporated

Service and maintenance technician (internship)

Austria

Aug 2018, Aug 2019

- **Robot maintenance:** SCARA robots and welding of robot components, including an articulated-arm gripper

Teaching and Leadership

UAS Technikum Wien

Team lead, Sumo-Bot competition

Vienna, Austria

2023 – 2025

- **Team lead:** guided a multidisciplinary team of seven through design, construction and programming of a competition robot; mentored members in programming, hardware design and troubleshooting

UAS Technikum Wien

Technical guide, open day

Vienna, Austria

2023 – 2025

- **Outreach:** guided prospective students through campus and laboratories and presented the degree programmes

Technical Skills

Robotics ROS/ROS 2, Gazebo, RViz, MoveIt, Nav2, behaviour trees, OPC UA, ABB RAPID/EGM, pymycobot

XR Unity, Meta XR SDK, OpenXR, hand tracking, passthrough, spatial anchors, digital twins

Languages C, C++, Python, Bash, JavaScript, TypeScript, SQL

Web and infrastructure React, Three.js, Next.js, Node.js, Docker, Git, CI/CD, AWS

Hardware ESP32/PlatformIO, Arduino, PCB design (EasyEDA), SolidWorks, Fusion 360, 3D printing, soldering

Vision and ML OpenCV, vision-language models, DeepFace, Whisper

Certificates

Dassault Systèmes, Mechanical Design Certificate, UAS Technikum Wien, 2023

Summer School “Quantum Technologies”, UAS Technikum Wien, 2023

Languages German (native) · English (fluent)